

EXERCISES AND EXACT SOLUTIONS

Teachers Where Learners Need Them

Independent practice, facilitated discussion, and assessment reference

Production Unit	PU-B07-C37
Notebook	V7_C_N02
Version	0.1
Assessment boundary	Learning evidence—not operational authorization

Instructions

Complete Part A before consulting the solutions. State assumptions explicitly. Where evidence is insufficient, write an abstention or escalation response rather than inventing precision.

Exercise 1

Add subject-specific shortages.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 2

Use full-time equivalents.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 3

Add housing/transport constraints.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 4

Explain why PTR improvement is not proof of learning impact.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exact solutions and scoring guide

Solution 1

Build school-subject-grade requirements and qualification-compatible supply.

Solution 2

Replace headcounts with contracted/available teaching time and document absence assumptions.

Solution 3

Model feasibility and costs without penalizing underserved schools.

Solution 4

Learning depends on pedagogy, attendance, materials, class composition, leadership, and implementation; causal impact needs evaluation.

Scoring rubric

Criterion	Strong evidence	Revision required
Decision framing	Owner, action, horizon, authority named	Model output substituted for decision
Evidence	Concepts, grain, provenance, limits explicit	Data treated as self-explanatory
Uncertainty	Abstention/escalation used appropriately	False precision or hidden limitations
Governance	Human authority and safeguards retained	Authority silently delegated